Category:

Network

Name:

N2: Silent call, Loud message

Message:

Description:

A VoIP call has been captured, but something about it seems… unusual. The signaling looks normal, and the call appears short and uneventful. However, the real message might not be in the SIP headers or the call metadata. Your task is to dig deeper into the media stream and uncover what’s truly being transmitted.

Details:

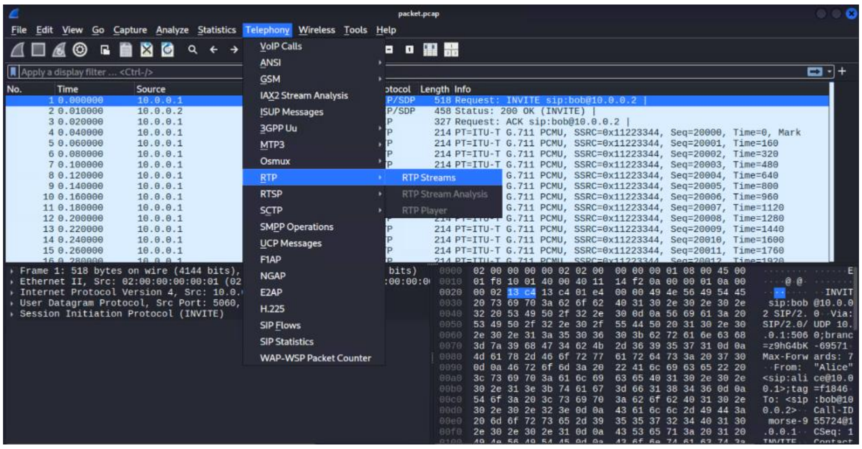
- You are given a packet capture file containing a VoIP session.
- The call uses standard protocols, but the payload may hold more than just “voice.”
- Analyze the signaling and media streams carefully. Identify the codec, extract the audio, and listen closely.
- The answer you seek is hidden in plain sound—decode it to reveal the flag.

Notes:

Flag format: CSG_FLAG{***the message you found***}

Instructions:

1. Open the packet.pcap by wireshark. Wireshark is pre-installed on the Kali attackbox.
2. Select Telephony -> RTP -> RTP Streams

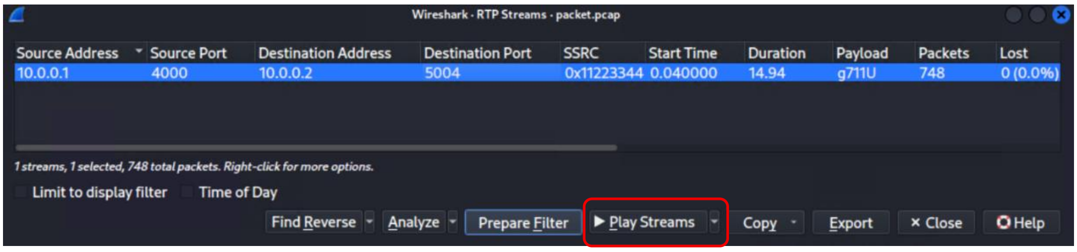

packet.pcap
File
Edit
View
Go
Capture
Analyze
Statistics
Telephony
Wireless
Tools
Help
Apply a display filter … <Ctrl-/>
VoIP Calls
ANSI
GSM
IAX2 Stream Analysis
ISUP Messages
3GPP Uu
MTP3
Osmux
RTP
RTSP
SCTP
SMPP Operations
UCP Messages
F1AP
NGAP
E2AP
H.225
SIP Flows
SIP Statistics
WAP-WSP Packet Counter
RTP Streams
RTP Stream Analysis
RTP Player

```
No.      Time       Source                      otocol   Length Info
       1 0.000000   10.0.0.1                    P/SDP    518 Request: INVITE sip:bob@10.0.0.2 |
       2 0.010000   10.0.0.2                    P/SDP    458 Status: 200 OK (INVITE) |
       3 0.020000   10.0.0.1                    P        327 Request: ACK sip:bob@10.0.0.2 |
       4 0.040000   10.0.0.1                    P        214 PT=ITU-T G.711 PCMU, SSRC=0x11223344, Seq=20000, Time=0, Mark
       5 0.060000   10.0.0.1                    P        214 PT=ITU-T G.711 PCMU, SSRC=0x11223344, Seq=20001, Time=160
       6 0.080000   10.0.0.1                    P        214 PT=ITU-T G.711 PCMU, SSRC=0x11223344, Seq=20002, Time=320
       7 0.100000   10.0.0.1                    P        214 PT=ITU-T G.711 PCMU, SSRC=0x11223344, Seq=20003, Time=480
       8 0.120000   10.0.0.1                                          G.711 PCMU, SSRC=0x11223344, Seq=20004, Time=640
       9 0.140000   10.0.0.1                                          G.711 PCMU, SSRC=0x11223344, Seq=20005, Time=800
      10 0.160000   10.0.0.1                                          G.711 PCMU, SSRC=0x11223344, Seq=20006, Time=960
      11 0.180000   10.0.0.1                                          G.711 PCMU, SSRC=0x11223344, Seq=20007, Time=1120
      12 0.200000   10.0.0.1                                          G.711 PCMU, SSRC=0x11223344, Seq=20008, Time=1280
      13 0.220000   10.0.0.1                    P        214 PT=ITU-T G.711 PCMU, SSRC=0x11223344, Seq=20009, Time=1440
      14 0.240000   10.0.0.1                    P        214 PT=ITU-T G.711 PCMU, SSRC=0x11223344, Seq=20010, Time=1600
      15 0.260000   10.0.0.1                    P        214 PT=ITU-T G.711 PCMU, SSRC=0x11223344, Seq=20011, Time=1760
```

```
› Frame 1: 518 bytes on wire (4144 bits),                   bits)
› Ethernet II, Src: 02:00:00:00:00:01 (02                  :00:00:0
› Internet Protocol Version 4, Src: 10.0.
› User Datagram Protocol, Src Port: 5060,
› Session Initiation Protocol (INVITE)
```

```
0000  02 00 00 00 00 02 02 00  00 00 00 01 08 00 45 00   ········ ······E
0010  01 f8 10 01 40 00 40 11  14 f2 0a 00 00 01 0a 00   ····@·@· ········
0020  00 02 13 c4 13 c4 01 e4  00 00 49 4e 56 49 54 45   ········ ··INVIT
0030  20 73 69 70 3a 62 6f 62  40 31 30 2e 30 2e 30 2e    sip:bob @10.0.0
0040  32 20 53 49 50 2f 32 2e  30 0d 0a 56 69 61 3a 20   2 SIP/2. 0··Via:
0050  53 49 50 2f 32 2e 30 2f  55 44 50 20 31 30 2e 30   SIP/2.0/ UDP 10.
0060  2e 30 2e 31 3a 35 30 36  30 3b 62 72 61 6e 63 68   .0.1:506 0;branc
0070  3d 7a 39 68 47 34 62 4b  2d 36 39 35 37 31 0d 0a   =z9hG4bK -69571·
0080  4d 61 78 2d 46 6f 72 77  61 72 64 73 3a 20 37 30   Max-Forw ards: 7
0090  0d 0a 46 72 6f 6d 3a 20  22 41 6c 69 63 65 22 20   ··From:  "Alice"
00a0  3c 73 69 70 3a 61 6c 69  63 65 40 31 30 2e 30 2e   <sip:ali ce@10.0
00b0  30 2e 31 3e 3b 74 61 67  3d 66 31 38 34 36 0d 0a   0.1>;tag =f1846·
00c0  54 6f 3a 20 3c 73 69 70  3a 62 6f 62 40 31 30 2e   To: <sip :bob@10
00d0  30 2e 30 2e 32 3e 0d 0a  43 61 6c 6c 2d 49 44 3a   0.0.2>·· Call-ID
00e0  20 6d 6f 72 73 65 2d 39  35 35 37 32 34 40 31 30    morse-9 55724@1
00f0  2e 30 2e 30 2e 31 0d 0a  43 53 65 71 3a 20 31 20   .0.0.1·· CSeq: 1
```

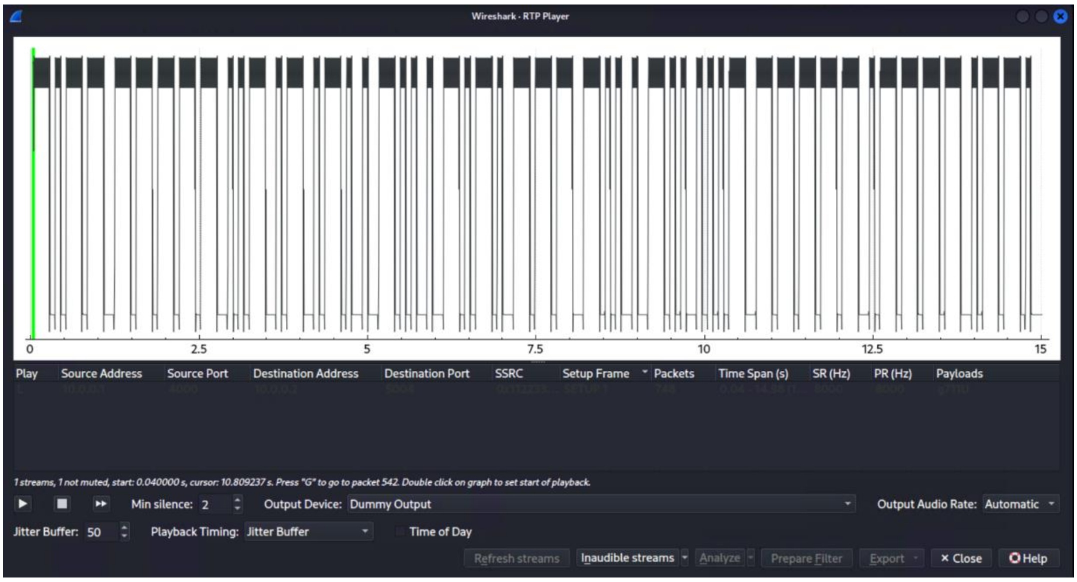
3. Select an RTP stream and Play Streams


Wireshark · RTP Streams · packet.pcap
Limit to display filter
Time of Day
Find Reverse
Analyze
Prepare Filter
▶ Play Streams
Copy
Export
× Close
Help

Source Address	Source Port	Destination Address	Destination Port	SSRC	Start Time	Duration	Payload	Packets	Lost
10.0.0.1	4000	10.0.0.2	5004	0x11223344	0.040000	14.94	g711U	748	0 (0.0%)

1 streams, 1 selected, 748 total packets. Right-click for more options.

4. Unfortunately, you can't play audio from Kali, but if you're sharp, you'll recognize this as Morse code just by looking at the audio graph.


Wireshark · RTP Player
0
2.5
5
7.5
10
12.5
15
Play
Source Address
Source Port
Destination Address
Destination Port
SSRC
Setup Frame
Packets
Time Span (s)
SR (Hz)
PR (Hz)
Payloads
Refresh streams
Inaudible streams
Analyze
Prepare Filter
Export
× Close
Help

1 streams, 1 not muted, start: 0.040000 s, cursor: 10.809237 s. Press "G" to go to packet 542. Double click on graph to set start of playback.

Min silence: 2
Output Device: Dummy Output
Output Audio Rate: Automatic
Jitter Buffer: 50
Playback Timing: Jitter Buffer
☐ Time of Day

5. Just decode the morse code:

–·–– ––––– ··– ·– ·–· · –·· · –·–· ––– –·· · –·· ···– ––––– ·–––– ·––·

Y 0 U ARE DE C O D E DV 0 1 P